

DATASCI 223 FINAL PROJECT

Cognitive Bias: Monolingual VS Bilingual

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Problem Statement

Bilingual patients tested in English on standard neuropsychological assessments may score differently than monolingual patients. This is not due to actual cognitive differences but rather due to the tests being designed and normed on English-speaking monolingual populations. As a result, this can lead to linguistic and cultural bias and then misdiagnosis of cognitive impairment in bilingual individuals in clinical settings. [1]

Two Main Questions:

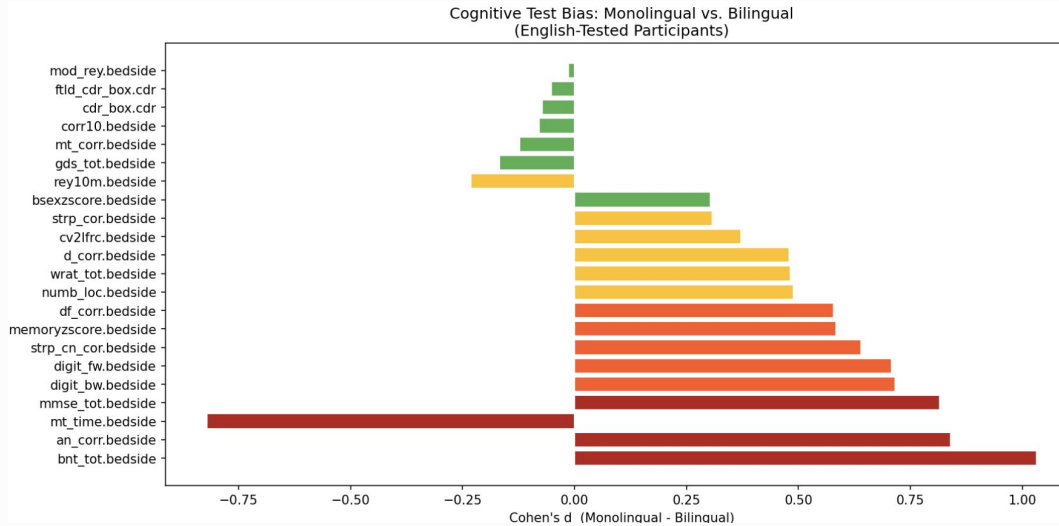
- Which cognitive tests show the largest performance gaps between monolingual and bilingual participants?
- Can a machine learning model tell them apart from test scores alone?

Where is the existing work pulled from?

- R analysis written for capstone research with Dr. de Leon at UCSF
 - Cohen's d bias analysis
 - Variable list
 - Cleaning logic
- Applied course methods
 - Classification pipeline structure
 - CV comparison workflow
 - ROC curves
 - confusion matrix
 - Feature importance



Phase 1: Bias Analysis



Of 22 tests analyzed:

- 4 showed large bias ($d \geq 0.8$)
- 5 showed medium bias
- 6 showed small bias
- 7 were fair

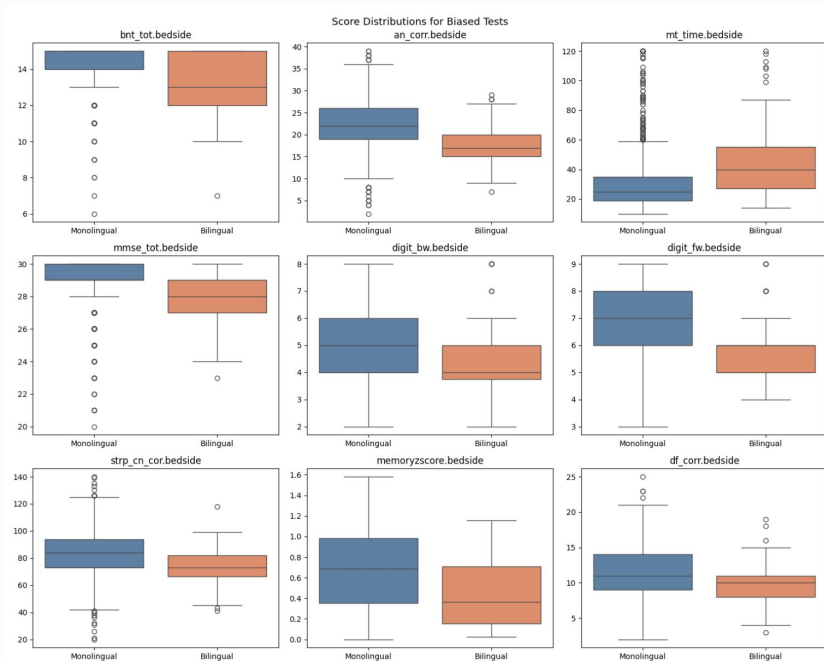
Green = Fair

Yellow = Small Bias

Orange = Medium Bias

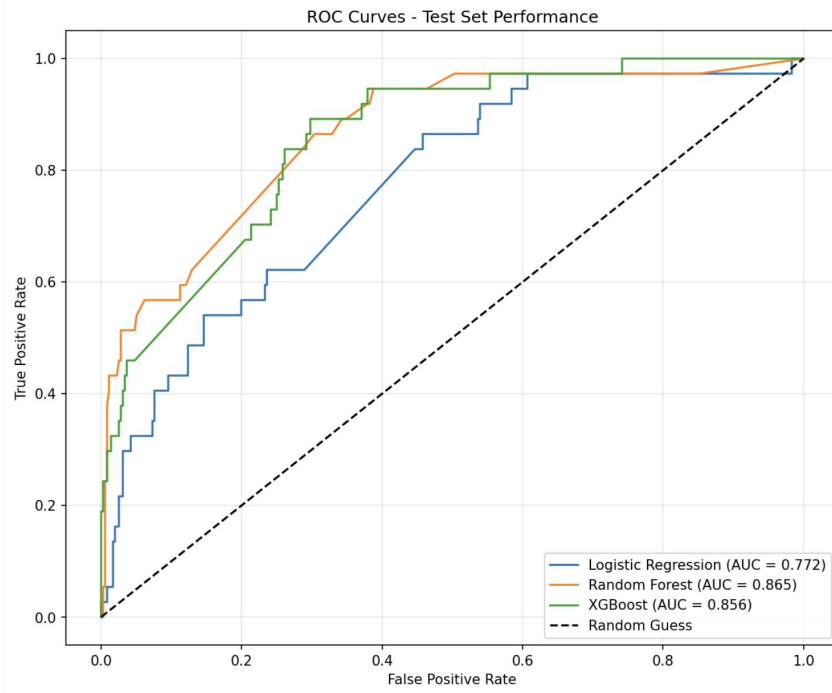
Red = Strong Bias

Phase 1: Bias Analysis



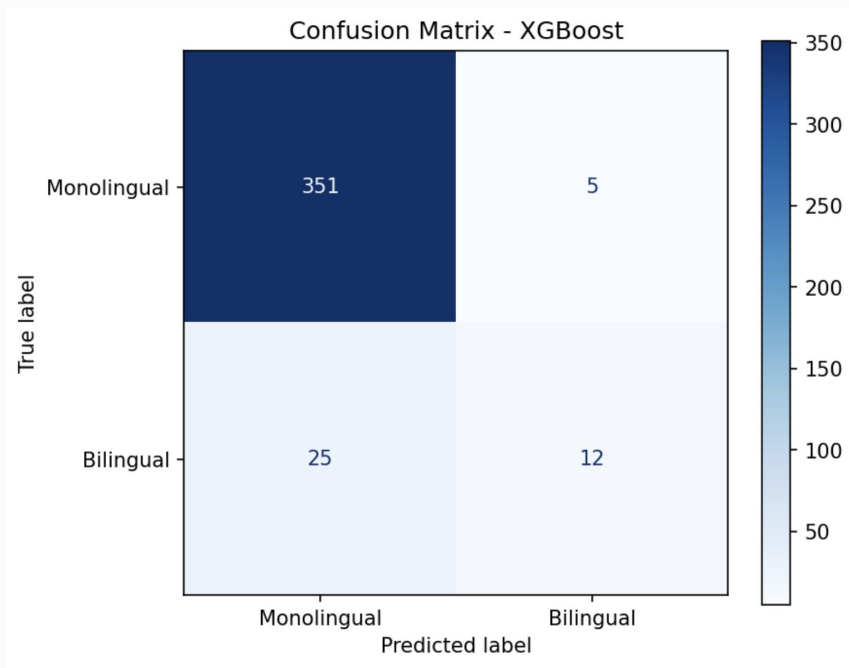
Score Distributions
for the medium
and large biased
tests

Phase 2: Classification



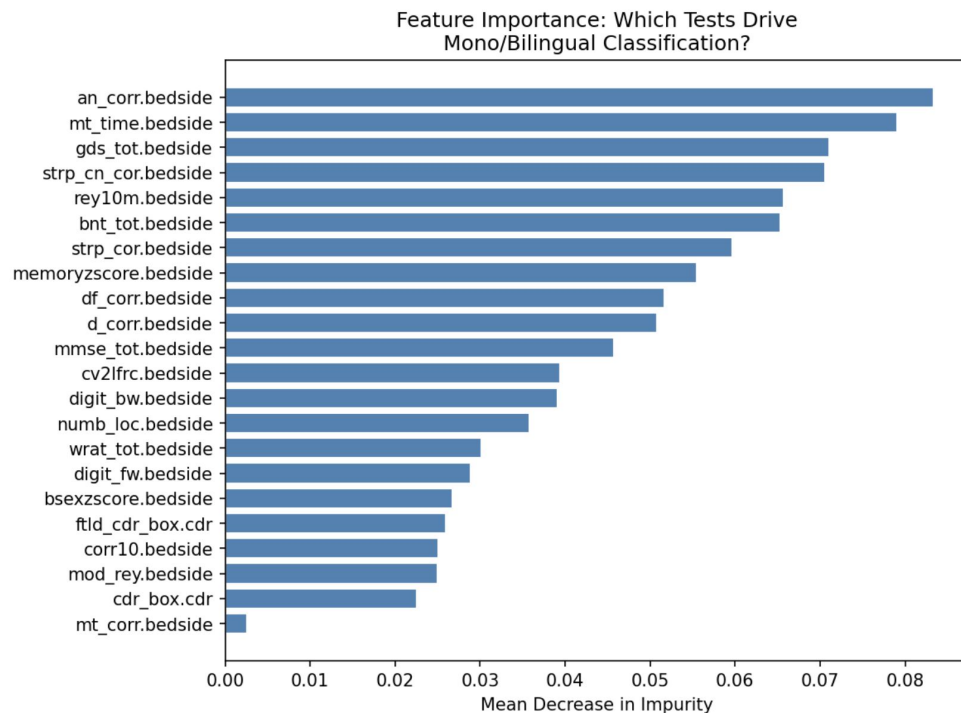
- Logistic Regression: AUC 0.716 (CV) / 0.772 (test)
- Random Forest: AUC 0.800 (CV) / 0.865 (test)
- XGBoost: AUC 0.821 (CV) / 0.856 (test)

Confusion Matrix



- Despite high overall accuracy (91-92%), bilingual recall was low across all models.
- The best model correctly identified only 12 of 37 bilingual participants in the test set

Feature Importance



- Animal Fluency and Modified Trails time were the strongest drivers of classification.
- The same tests that showed the largest Cohen's d effect sizes in Phase 1
- Confirms that the tests identified as most biased are also the ones the model relies on most to tell the two groups apart

Contributions

- Translated existing R bias analysis into Python
- Extended the analysis with supervised ML classification
 - Asked whether a model could predict group membership from test scores alone
- Connected the two findings
 - The tests with the highest bias also had the highest feature importance in the classifier



Tools/Methods Used

- Data cleaning and statistical analysis
 - pandas, numpy, scipy
- Effect size for bias quantification
 - Cohen's d
- Scikit-learn
 - train/test split, StandardScaler, SimpleImputer, StratifiedKFold, Logistic Regression, Random Forest
- XGBoost
- matplotlib
- seaborn



Issues Overcame

- No participant had complete data across all 22 tests. Dropping missing values left zero samples and required median imputation
- Severe class imbalance (10:1) meant high accuracy was misleading. Bilingual recall remained low even in the best model
- Negative values in the dataset were missing data codes, not real scores. I had to identify and replace them before any analysis



Conclusion

- 15 of 22 neuropsychological tests showed measurable bias between monolingual and bilingual participants
- All three classifiers performed well above chance with a best test AUC of 0.865
- The tests with the highest bias also had the highest feature importance which confirms the bias is systematic and learnable
- Future work: apply SMOTE to address class imbalance and improve bilingual recall



Citations

1. Lehtonen, M., Soveri, A., Laine, A., Järvenpää, J., de Bruin, A., & Antfolk, J. (2018). Is bilingualism associated with enhanced executive functioning in adults? A meta-analytic review. *Psychological Bulletin*, 144(4), 394–425. <https://doi.org/10.1037/bul0000142>
2. Malvin, D. (2025). Cognitive test bias analysis: Monolingual vs. Bilingual. de Leon Lab, UCSF.